

FOREWORD

You are now the proud owner of a KUBOTA Diesel Engine. This KUBOTA Diesel Engine is a product of quality engineering and manufacturing. It is made of fine materials and under a rigid quality control system. Over the year, KUBOTA has pledged to always improve on its products. This enduring commitment saw KUBOTA accredited with compulsory Thai Industrial Standards (TIS) from the Ministry of Industry as well as certifications from the International Organization for Standardization on Quality Management System (ISO 9001: 2000) and Environmental Management System (ISO 14001). These are genuine testaments to the excellent quality of the KUBOTA Diesel Engine that will secure you a long and satisfactory service.

To obtain the best use of your KUBOTA Diesel Engine, please read this manual carefully. It will help you become familiar with the operation of the KUBOTA Diesel Engine and contains many helpful hints about the KUBOTA Diesel Engine maintenance. For detailed information or inquiries, please do not hesitate to consult with the manufacturer or authorized dealers.

KUBOTA reserves the right to change specifications, technical details, or other information in regards to the product or related equipment without prior notice.



CERT.NO. 403581

Quality Management System
(ISO9001:2000)



ISO 14001
EMS99001/001

Environmental Management
System
(ISO14001)



มอก.787-2551

Compulsory
Thai Industrial Standards
(TIS)

SAFETY FIRST

This symbol, the industry's "Safety Alert Symbol", is used throughout this manual and on safety labels on the machine itself to warn of the possibility of personal injury or death. Read these instructions carefully. It is essential that you read and follow all instructions and safety regulations before assembling or operating this KUBOTA Diesel Engine.

 **DANGER** : Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.

 **WARNING** : Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

 **CAUTION** : Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

■ **IMPORTANT** : Indicates that equipment or property damage could result if instructions are not followed.

■ **NOTE** : Gives helpful information.

KUBOTA LIMITED WARRANTY

Upon the purchase of the KUBOTA Diesel Engine, please fill in the warranty registration card and return it to the dealer for a 3 year limited warranty. The warranty covers the KUBOTA Diesel Engine whose damages originate from defective parts and/or workmanship. This warranty is null and void under the following circumstances:

1. Damages that occur from negligence of the maintenance specified in the Operation and Maintenance Manual.
2. Damages that occur from operations which are incorrect, improper, or not compliance with the intended purpose as specified in the Operation and Maintenance Manual, and improper maintenance of the KUBOTA Diesel Engine.
3. Damages that occur from accidents or natural calamities.
4. Damages that occur from improper modification and repairing.
5. Damages that occur from the maintenance not serviced by KUBOTA authorized dealers.
6. Damages that occur from using non-genuine parts, unqualified or low-quality lubricants, or those that do not meet the specifications in the Operation and Maintenance Manual of the KUBOTA Diesel Engine.
7. Damages, wears, or rusts of the exterior parts from vibration and friction.
8. Noise or vibration which does not affect the performance of the KUBOTA Diesel Engine.

The company shall not be liable for any consequential damage arising during the period in which the diesel engine is damaged or in repair. If this product is exported outside of the country or its territories, the warranty is void. The warranty card cannot be reissued if lost.

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SAFE OPERATION

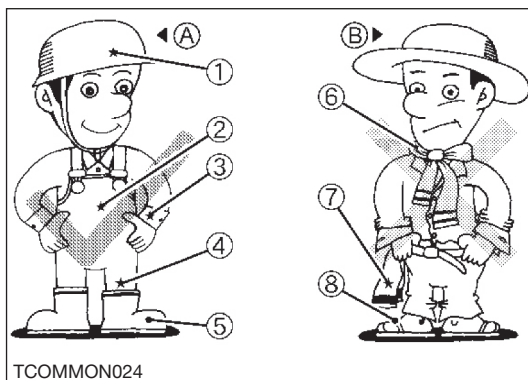
Read and understand this manual carefully before operating the KUBOTA Diesel Engine or any implements attached to it. This part contains important precautions which operator should follow for safe operation as below.

BEFORE OPERATING THE KUBOTA DIESEL ENGINE

1. Read and understand the Operation Manual of the KUBOTA Diesel Engine before operating. Lack of proper knowledge may lead to accidents.
2. It is the owner's responsibility to ensure that anyone who will operate the KUBOTA Diesel Engine read this manual first and becomes familiar with safe operation.
3. Visually check for abnormal, missing, or malfunctioning parts. Make necessary repairs before operating.
4. Replace damaged or illegible safety labels.
5. Assemble, remove, and install the KUBOTA Diesel Engine according to the instructions contained in this manual only. Operation other than these may result in fault which may lead to serious injury or death.

DURING OPERATION OF THE KUBOTA DIESEL ENGINE

1. Keep children and livestock away when operating the KUBOTA Diesel Engine.
2. When operating or being near the KUBOTA Diesel Engine, do not wear loose or tattered cloth as it may lead to potential hazards if caught in rotating parts. It is recommended to wear protective equipment such as a safety helmet, safety shoes, boots, protective goggles, earplugs, or protective gloves, etc.



(A) Good example

- (1) *Safety helmet*
- (2) *Overall or protective wear*
- (3) *Tightly fit sleeves*
- (4) *Neatly fit pants bottoms*
- (5) *Anti-slip shoes*

(B) Bad example

- (6) *Neck towel*
- (7) *Waist towel*
- (8) *Sandals*

AFTER OPERATING THE KUBOTA DIESEL ENGINE

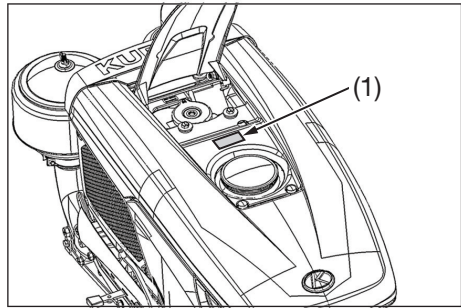
1. Store the KUBOTA Diesel Engine on a level surface after use.
2. When detaching the KUBOTA Diesel Engine from other implements, make sure to place the detached engine on a hard and level surface.

MAINTENANCE OF THE KUBOTA DIESEL ENGINE

1. Always wear proper protective equipment when servicing the machine.
2. Do not modify the KUBOTA Diesel Engine. Unauthorized modification may affect the function of the KUBOTA Diesel Engine, which may cause danger to the operator.
3. Do not perform maintenance on the KUBOTA Diesel Engine during engine operation as it may cause danger to the operator.

SAFETY LABELS

(1) Part no. 1WA71-L14A△



T20ZTA001-TH

Care of the danger, caution, and warning labels

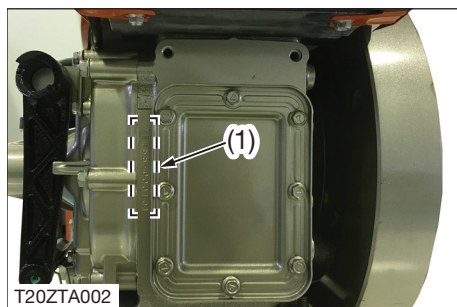
1. Keep the safety labels clean and free from obstructing material.
2. Clean the safety labels with soap and water, then dry with a soft cloth.
3. Replace the damaged or missing safety label with a new one from your local KUBOTA authorized dealer.
4. If the original safety label is replaced with a new one, make sure the new label is placed on the same location.
5. Attach the new safety label on a clean dry surface. Press to expel all bubbles from the label before attaching.

AFTER SALES SERVICE

The authorized dealer of this KUBOTA Diesel Engine is ready to offer their services to help you get the most out of the KUBOTA Diesel Engine.

After reading the Operation and Maintenance Manual thoroughly, you will find that some of the regular maintenance can be performed by yourself.

However, when in need of parts or repair service, please contact your local authorized dealer, and provide your dealer with the serial number (1) of the KUBOTA Diesel Engine around the location as in the photo.



(1) Serial number

Please fill in the details below.

KUBOTA Diesel Engine

Model Name:

Serial Number:.....

Date of Purchase:.....

KUBOTA Dealer:

SPECIFICATIONS

SPECIFICATION TABLE

Pre-combustion Swirl Chamber (KUBOTA TVCS: Three Vortex Combustion System)

Model of the KUBOTA Diesel Engine		ZT100		ZT110		ZT120	
		Standard	Pro	Standard	Pro	Standard	Pro
Type		Water-cooled 4-stroke horizontal diesel engine					
No. of cylinder	Unit	1					
Cylinder bore x stroke	mm	88 x 90		92 x 90		94 x 90	
Displacement volume	cm³	547		598		624	
Maximum power	HP/rpm (kW/rpm)	10.0/2400 (7.40/2400)		11.0/2400 (8.10/2400)		12.0/2400 (8.83/2400)	
Continuous rated power	HP/rpm (kW/rpm)	9.0/2400 (6.60/2400)		9.5/2400 (7.00/2400)		10.5/2400 (7.72/2400)	
Specific fuel consumption at continuous rated power	g/kW-h	269					
Compression ratio		22.0:1		20.5:1		21.0:1	
Valve clearance	mm	0.195 - 0.235					
Maximum torque	kgf-m/rpm (N-m/rpm)	3.6/1600 (35.3/1600)		4.0/1600 (39.2/1600)		4.3/1600 (42.2/1600)	
Cooling water capacity	Liter	2.2					
Fuel tank capacity	Liter	10.9					
Lubricating oil capacity	Liter	2.8					
Fuel type		Automotive Diesel Oil (SAE No.2 - D), Biodiesel fuel (B7~B20)					
Lubricating oil type		SAE 10W-30, SAE 10W-40					
Combustion system type		Pre-combustion swirl chamber (KUBOTA TVCS: Three vortex combustion system/special design pre combustion swirl chamber with indirect injection: IDI)					
Cooling system type		Pressurized radiator					
Lubricating system type		Trochoid pump					
Air cleaner type		Wet-type (Oil bath type)					
Starting system type		Double speed hand starting / ES models feature an electric starter					
LED headlamp		✓	✓	✓	✓	✓	✓
LED turn signal		-	✓	-	✓	-	✓
USB charging unit		△	✓	△	✓	△	✓

Model of the KUBOTA Diesel Engine	ZT100		ZT110		ZT120	
	Standard	Pro	Standard	Pro	Standard	Pro
Key switch starter	✓ (ZT-SW model only)					
Dimensions	mm	369 x 802 x 559			369 x 819 x 559	
Weight						
Hand starter model	kg	105	110		111	
Electric starter model (ZT-ES)	kg	112			113	
Key switch starter model (ZT-SW)	kg	113			114	

✓: Standard accessory △: Optional accessory

NOTE:

1. The values shown are estimated by the manufacturer and are subject to change at any time without prior notice.
2. Horsepower, torque, and specific fuel consumption are values from the engine which operates for at least 50 hours at rated rpm, and in compliance to JIS B 8018 at standard environmental conditions. (Atmospheric pressure: 750 mmHg, temperature: 25 degrees Celsius, RH: 30%)

Direct Injection (DI)

Model of the KUBOTA Diesel Engine		ZT100DI		ZT110DI		ZT125DI		ZT140DI		ZT155DI	
		Standard	Pro	Standard	Pro	Standard	Pro	Standard	Pro	Standard	Pro
Type		Water-cooled 4-stroke horizontal diesel engine									
No. of cylinder	Unit	1									
Cylinder bore x stroke	mm	88 x 90		92 x 90		94 x 96		97 x 96		100 x 98	
Displacement volume	cm³	547		598		666		709		770	
Maximum power	HP/rpm (kW/rpm)	10.0/2400 (7.35/2400)		11.0/2400 (8.09/2400)		12.5/2400 (9.19/2400)		14.0/2400 (10.3/2400)		15.5/2400 (11.4/2400)	
Continuous rated power	HP/rpm (kW/rpm)	9.0/2400 (6.62/2400)		9.5/2400 (6.99/2400)		11.0/2400 (8.09/2400)		12.5/2400 (9.19/2400)		13.5/2400 (9.93/2400)	
Specific fuel consumption at continuous rated power	g/kW·h	240									
Compression ratio		18.0:1								17.5:1	
Valve clearance	mm	0.195 - 0.235									
Maximum torque	kgf·m/rpm (N·m/rpm)	3.4/1600 (33.4/1600)		3.8/1600 (37.3/1600)		4.7/1600 (46.10/1600)		5.0/1600 (49.00/1600)		5.5/1600 (53.94/1600)	
Cooling water capacity	Liter	2.1									
Fuel tank capacity	Liter	10.9									
Lubricating oil capacity	Liter	2.8									
Fuel type		Automotive Diesel Oil (SAE No.2 - D), Biodiesel fuel (B7~B20)									
Lubricating oil type		SAE 10W-30, SAE 10W-40									
Combustion system type		Direct Injection (DI)									
Cooling system type		Pressurized radiator									
Lubricating system type		Trochoid pump									
Air cleaner type		Wet-type (Oil bath type)									
Starting system type		Double speed hand starter / ES models feature an electric starter									
LED headlamp		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LED turn signal		-	✓	-	✓	-	✓	-	✓	-	✓
USB charging unit		△	✓	△	✓	△	✓	△	✓	△	✓
Key switch starter		✓ (ZT-SW model only)									
Dimensions	mm	369 x 819 x 559									

Model of the KUBOTA Diesel Engine		ZT100DI		ZT110DI		ZT125DI		ZT140DI		ZT155DI	
		Standard	Pro	Standard	Pro	Standard	Pro	Standard	Pro	Standard	Pro
Weight											
Hand starter model	kg	106		111				112			
Electric starter model (ZT-ES)	kg	113						114			
Key switch starter model (ZT-SW)	kg	114						115			

✓: Standard accessory △: Optional accessory

NOTE:

1. The values shown are estimated by the manufacturer and are subject to change at any time without prior notice.
2. Horsepower, torque, and specific fuel consumption are values from the engine which operates for at least 50 hours at rated rpm, and in compliance to JIS B 8018 at standard environmental conditions. (Atmospheric pressure: 750 mmHg, temperature: 25 degrees Celsius, RH: 30%)

Model of the KUBOTA Diesel Engine		ZT180DI	ZT180DI-ESE
		Standard	Standard
Type		Water-cooled 4-stroke horizontal diesel engine	
No. of cylinder	Unit	1	
Cylinder bore x stroke	mm	107 x 105	
Displacement volume	cm ³	944	
Maximum power	HP/rpm (kW/rpm)	18.0/2400 (13.4/2400)	
Continuous rated power	HP/rpm (kW/rpm)	16.0/2400 (11.9/2400)	
Specific fuel consumption at continuous rated power	g/kW-h	240	
Compression ratio		18.2:1	
Valve clearance	mm	0.195 - 0.235	
Maximum torque	(kgf·m/rpm) (N·m/rpm)	6.5/1600 (63.74/1600)	
Cooling water capacity	Liter	2.9	
Fuel capacity	Liter	14.5	
Lubricating oil capacity	Liter	3.7	
Fuel type		Automotive Diesel Oil (SAE No.2 - D), Biodiesel fuel (B7~B20)	
Lubricating oil type		SAE 10W-30, SAE 10W-40	
Combustion system type		Direct Injection (DI)	
Cooling system type		Pressurized radiator	
Lubricating system type		Trochoid pump	
Air cleaner type		Wet type (Oil bath type)	
Starting system type		Double speed hand starting / ES models feature an electric starter system	
LED headlamp		✓	-
LED turn signal		-	
USB charging unit		△	-
Key switch starter		✓ (ZT180DI-SW model only)	-
Dimensions	mm	405 x 887 x 644	

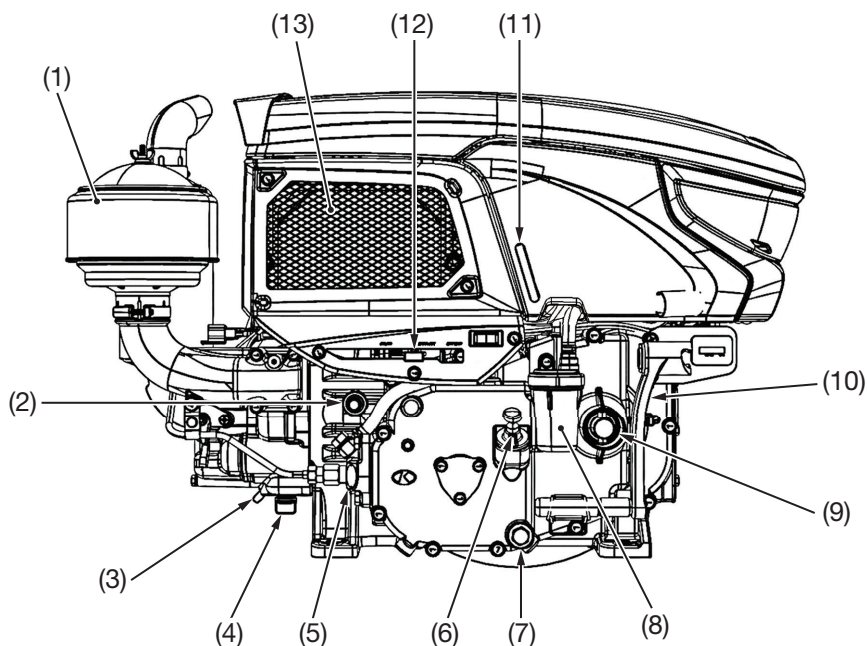
Model of the KUBOTA Diesel Engine	ZT180DI		ZT180DI-ESE	
	Standard		Standard	
Weight				
Hand starter model	kg	145		
Electric starter model (ZT-ES, ZT-ESE)	kg	150		
Key switch starter model (ZT-SW)	kg	151		

✓: Standard accessory △: Optional accessory

- NOTE:**
1. The values shown are estimated by the manufacturer and are subject to change at any time without prior notice.
 2. Horsepower, torque, and specific fuel consumption are values from the engine which operates for at least 50 hours at rated rpm, and in compliance to JIS B 8018 at standard environmental conditions. (Atmospheric pressure: 750 mmHg, temperature: 25 degrees Celsius, RH: 30%)

PARTS NAME

DIESEL ENGINE PARTS



(1) *Wet-type air cleaner*

(2) *Engine oil pressure gauge*

(3) *Decompression lever*

(4) *Water drain plug*

(5) *Fuel injection pump*

(6) *Engine oil dipstick*

(7) *Engine oil filter and drain plug*

(8) *Fuel filter*

(9) *Starting shaft*

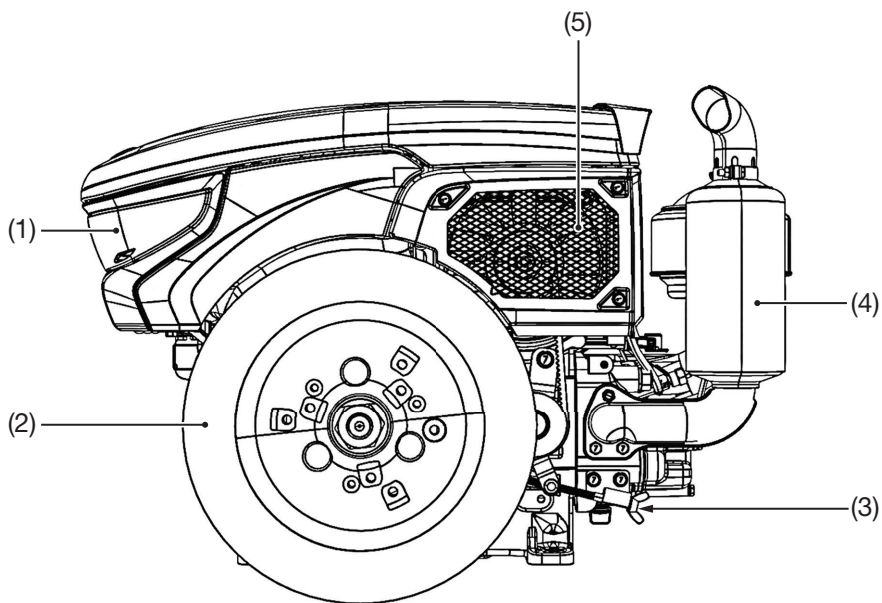
(10) *Starting handle*

(11) *Fuel gauge*

(12) *Accelerator lever*

(13) *Radiator*

NOTE: This illustration of parts is for one model only. For other models, the shape may vary.



T20ZTA004

(1) *Headlamp and turn signal
(Pro model only)*

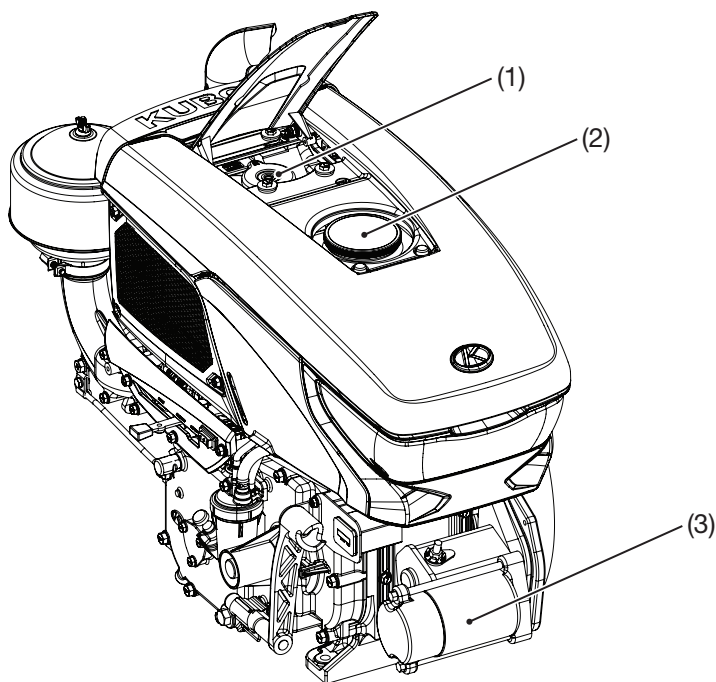
(2) *Flywheel*

(3) *Belt tension adjusting wing nut*

(4) *Muffler*

(5) *Cooling fan*

NOTE: This illustration of parts is for one model only. For other models, the shape may vary.



T20ZTA005

(1) *Radiator cap*

(2) *Fuel tank cap*

(3) *Starter motor*

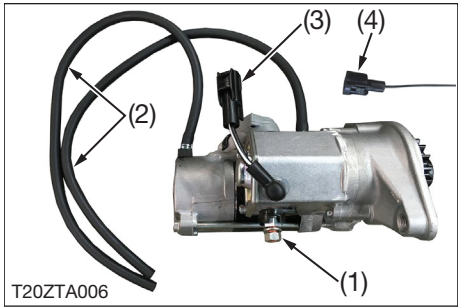
(Electric starter model only)

NOTE: This illustration of parts is for one model only. For other models, the shape may vary.

ADDITIONAL PARTS FOR THE ELECTRIC STARTER MODEL

Parts furnished with the engine

- 1. The starter motor includes B terminal (1), breather hoses (specific models only) (2), ST terminal (3), and plug for connecting wires (4).



- (1) *B terminal*
- (2) *Breather hoses (specific models only)*
- (3) *ST terminal*
- (4) *Plug for connecting wires*

Parts to be procured at site

No.	Part list	Amount
1	#2 wire (2 mm ²)	1 units
2	#3 wire (3 mm ²)	1 units
3	#25 wire (25 mm ²)	2 units
4	Fuse (30A)	1 unit
5	12V battery (minimum 30A)	1 unit
6	3 or 4 position starter switch	1 unit

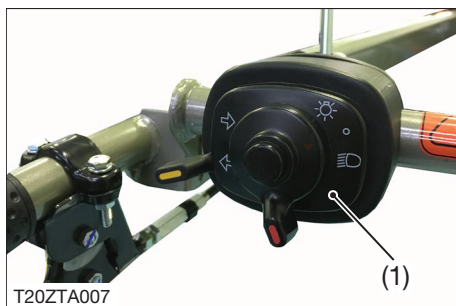
■ NOTE

- The size of the battery depends on the power requirement of the implement attached to the engine.

INSTALLING ACCESSORIES

INSTALLING THE HEADLAMP AND TURN SIGNAL SWITCH TO THE POWER TILLER (PRO MODEL ONLY)

1. Install the headlamp and turn signal switch (1) to the left handle fixing bar of the power tiller and tighten the nut.



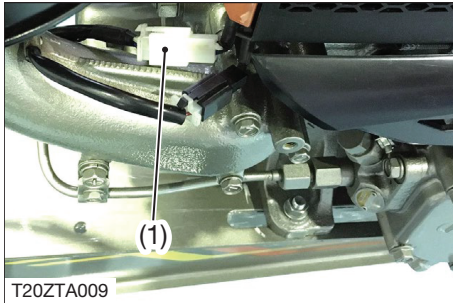
(1) Headlamp and turn signal switch

2. Tuck the wires neatly inside the power tiller handle (1). Open the top cover and the tool tray (2) to allow the wire to pass over the gear case to the right, then the back of the engine.



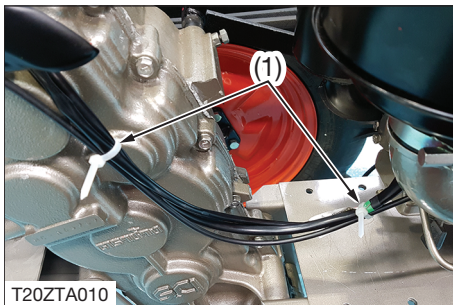
(1) Power tiller handle
(2) Top cover and tool tray
--- Wiring direction

3. Connect the wiring connectors (1).



(1) Headlamp and turn signal switch connectors

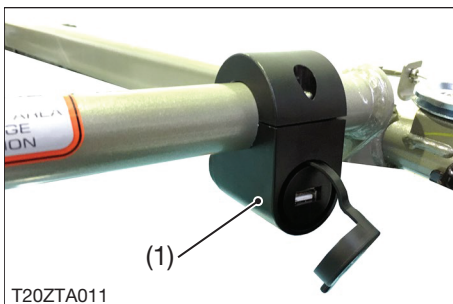
4. Fix the wires with the provided cable ties (1) at 2 points as in the photo.



(1) Cable ties

CONNECTING THE USB CHARGING UNIT TO THE POWER TILLER

1. Install the USB charging unit (1) on the right handle fixing bar of the power tiller and tighten the nut.



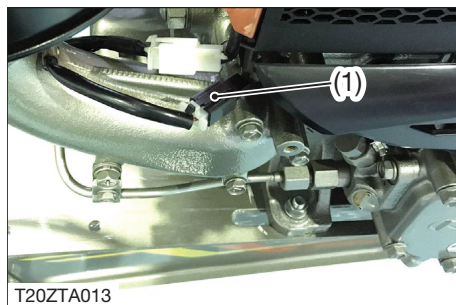
(1) USB charging unit

2. Tuck the wires neatly inside the power tiller handle (1). Open the top cover and the tool tray (2) to allow the wire to pass over the gear case to the back of the engine.



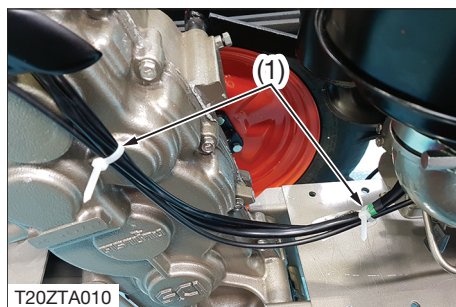
- (1) Power tiller handle
- (2) Top cover and tool tray
- Wiring direction

3. Connect the wiring connector (1).



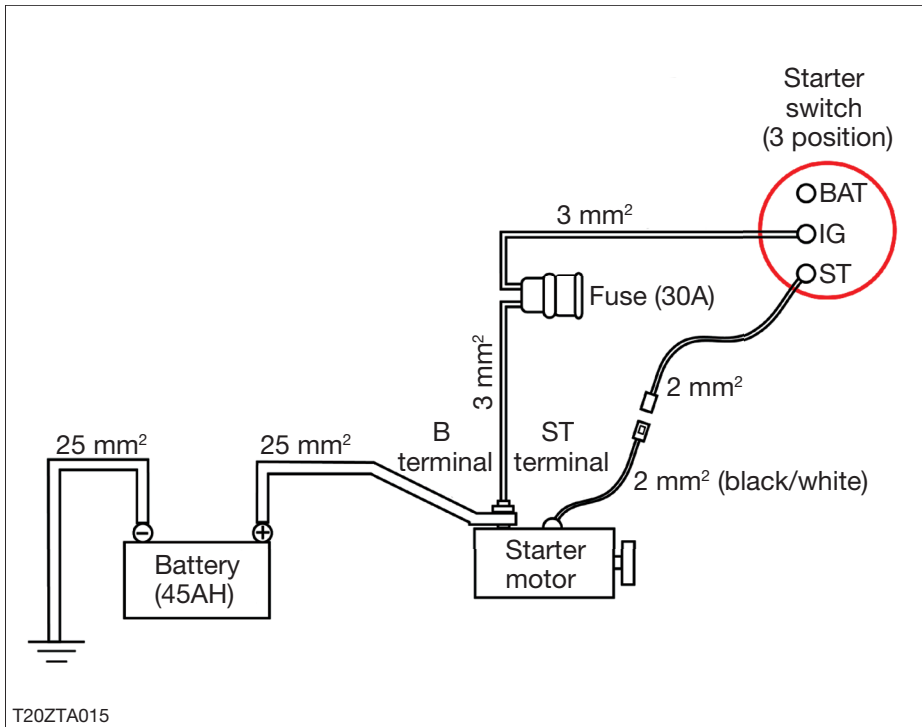
- (1) USB charging unit connector

4. Fix the wires with the provided cable ties (1) at 2 points as in the photo.



- (1) Cable ties

INSTALLING THE ELECTRIC STARTER (ELECTRIC STARTER MODEL ONLY)



1. Wiring from the battery

- Connect a 25 mm² cable from the battery's “negative (-) terminal” to the ground, attaching it to the metal frames of the implement.
- Connect a 25 mm² cable from the battery's “positive (+) terminal” to the “B terminal” on the starter motor.

2. Wiring from the starter motor

- Connect the #3 wire from the motor starter's “B terminal” through the fuse (30A) and to the “IG terminal” of the starter switch.
- Install the plug on the starter motor's “ST terminal”, then connect the #2 wire to the “ST terminal” of the starting motor.

BEFORE FIRST TIME USE

CAUTION

To avoid personal injury or engine damage:

- **Strictly comply with the following instructions. Otherwise, the diesel engine may be severely damaged.**

The engine will not be filled up with fluids, lubricants, and fuel from the factory. Therefore, the operator should perform the following operations before the first-time operation:

1. Fill up the radiator with clean water (see page 25).
2. Use only diesel oil (see page 26).
3. Fill the engine oil to the marked level (see page 27).
4. At the wet-type air cleaner, fill up the air cleaner oil pan with the engine oil to the marked level (see page 29).
5. Check the fan belt tension (see page 31).

CHECKING AND ADJUSTING BEFORE THE OPERATION

⚠ CAUTION

To avoid personal injury or engine damage:

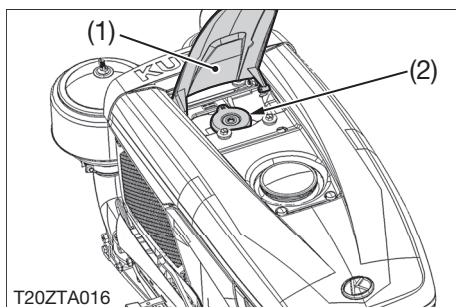
- When performing check before the operation, always wear protective equipment and strictly follow the procedures.

CHECKING THE COOLING WATER LEVEL

⚠ CAUTION

To avoid personal injury or engine damage:

- Before removing the radiator cap to check the cooling water level, always wait until the engine cooled down sufficiently to prevent injury from high-temperature water in the radiator.
1. Lift up the hood (1), then remove the radiator cap (2) by turning.



- (1) Hood
(2) Radiator cap

2. Check if the water is at the marked level. It must be above the radiator core.
3. If the water is below the marked level, fill up with clean water (e.g. tap water) through the water fill port.

Radiator cooling water capacity	ZT100, ZT110, ZT120	2.2 liters
	ZT100DI, ZT110DI, ZT125DI, ZT140DI, ZT155DI	2.1 liters
	ZT180DI	2.9 liters

■ IMPORTANT

- Only use clean water to fill in the radiator. Do not use unclean water, or the engine may overheat due to the radiator clogging.
4. Re-tighten the radiator cap firmly to prevent water from spilling out of the radiator during operation.

CHECKING THE FUEL LEVEL

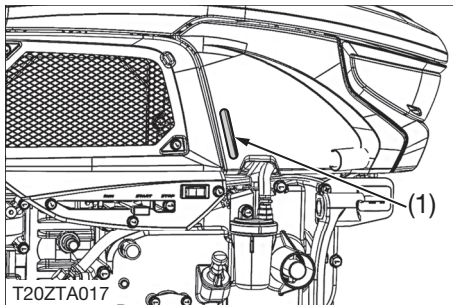
⚠ CAUTION

To avoid personal injury or engine damage:

- Check and re-fill fuel in a well-ventilated area and away from spark to prevent fire.

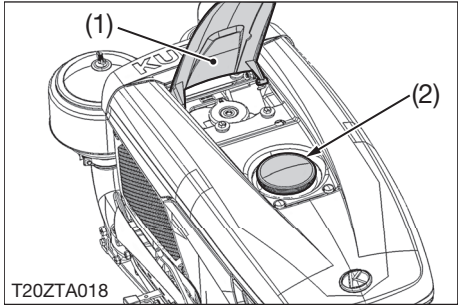
■ IMPORTANT

- Always re-fill the fuel tank after operation to prevent water vapor from forming within the tank. Wipe off any oil spills.
 - Use diesel oil only.
1. Always check the fuel level at the fuel gauge (1) before use.



(1) Fuel gauge

2. If the fuel level is low, lift up the hood (1), then remove the fuel tank cap (2) by turning.



- (1) Hood
(2) Fuel tank cap

3. Always fill up the fuel tank at the fuel fill port before use.

Fuel type		Automotive Diesel Oil (SAE No.2 - D), Biodiesel fuel (B7~B20)
Fuel capacity	ZT100, ZT110, ZT100DI, ZT110DI, ZT120, ZT125DI, ZT140DI, ZT155DI	10.9 liters
	ZT180DI	14.5 liters

4. Re-tighten the fuel tank cap firmly to prevent fuel from spilling out of the tank during operation.

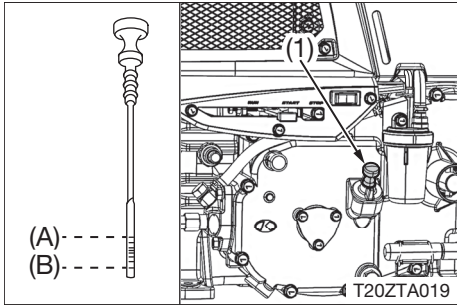
CHECKING THE ENGINE OIL LEVEL AND FILLING THE ENGINE OIL

CAUTION

To avoid personal injury or engine damage:

- Do not pull the engine oil dipstick when the engine is running.
1. Set the engine on ground level

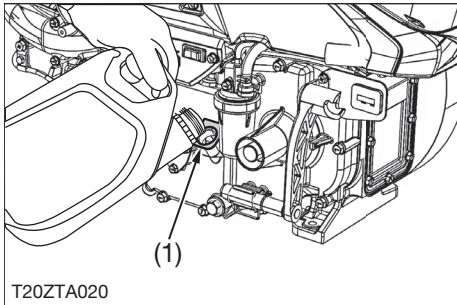
2. Pull the engine oil dipstick (1) out and clean. Then insert it back in place and pull out again. Check if the engine oil level is between the upper mark (A) and lower mark (B).



(1) Engine oil dipstick
(A) Upper mark
(B) Lower mark

3. If the engine oil level is above the upper mark, drain the oil (see page 48), then check until it is between the upper mark and lower mark.
4. If the engine oil level is below the lower mark, fill the fuel at the engine oil fill port (1), then check until it is between the upper mark and lower mark.

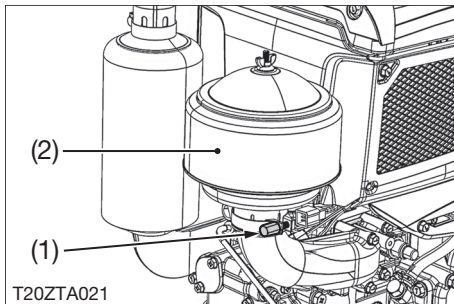
Engine oil type		TRA CHANG Engine Oil SAE 10W-30, SAE 10W-40
Engine oil capacity	ZT100, ZT110, ZT100DI, ZT110DI, ZT120, ZT125DI, ZT140DI, ZT155DI	2.8 liters
	ZT180DI	3.7 liters



(1) Engine oil fill port

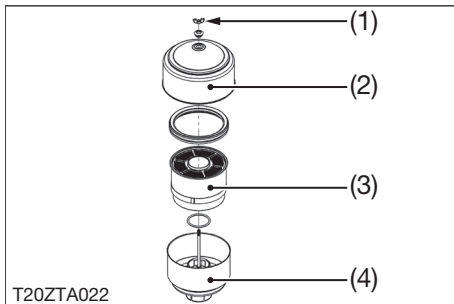
CHECKING THE AIR CLEANER (WET TYPE)

1. Loosen the clamp bolt (1) and remove the air cleaner (2) from the engine.



- (1) *Clamp bolt*
- (2) *Air cleaner*

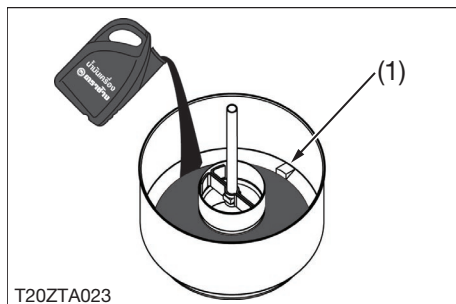
2. Loosen the wing nut (1) and remove the air cleaner cover (2).
3. Pull out the air cleaner filter (3) from the oil pan (4).



- (1) *Wing nut*
- (2) *Air cleaner cover*
- (3) *Air cleaner filter*
- (4) *Air cleaner oil pan*

4. Check the oil level in the air cleaner, and check if the air cleaner filter is dirty or damaged.
5. If it is dirty, drain the oil. Then clean the air cleaner filter and oil pan with fuel.

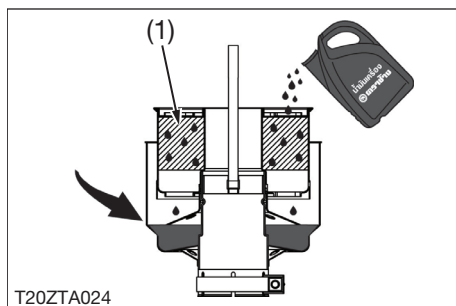
6. Re-fill the engine oil in the oil pan to the marked level (1).



(1) *Marked level*

7. Soak the top of air cleaner filter (1) with engine oil to optimize filtering efficiency.

Engine oil in the oil pan	TRA CHANG Engine Oil SAE 10W-30, SAE 10W-40
---------------------------	--



(1) *Air cleaner filter*

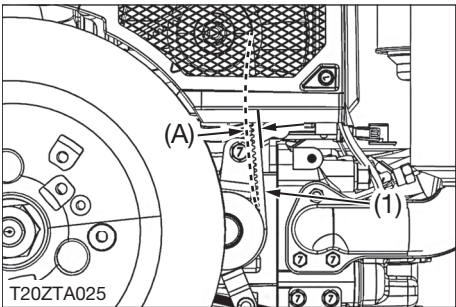
■ NOTE

- Engine oil to be filled in the air cleaner oil pan should be the same type as in the engine.

CHECKING AND ADJUSTING THE FAN BELT TENSION

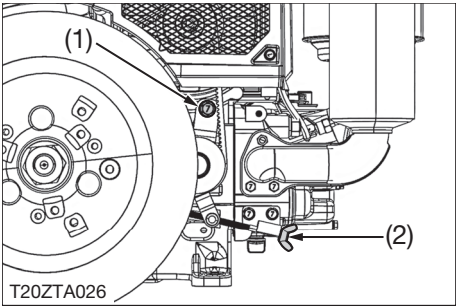
- 1. Use one finger to press down the middle of the fan belt (1) to check whether the belt deflection (A) is within the specified value.

Fan belt tension	ZT100, ZT100DI, ZT110, ZT110DI, ZT120, ZT125DI, ZT140DI, ZT155DI	11 to 15 mm 0.44 to 0.59 in
	ZT180DI, ZT180DI-ESE	18 to 22 mm 0.71 to 0.86 in



- (1) Fan belt
- (A) Belt deflection

- 2. If the belt deflection is out of the specified value, loosen the tension pulley bolt (1).
- 3. Adjust the wing nut (2) until the belt tension is within the specified value.



- (1) Tension pulley bolt
- (2) Wing nut

- 4. Re-tighten the tension pulley bolt (1).

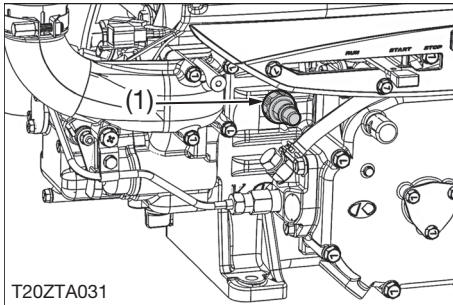
OPERATING THE DIESEL ENGINE

STARTING AND STOPPING THE ENGINE

Engine with starting handle

How to check the engine oil pressure gauge

1. Before the engine starts, the engine oil pressure gauge (1) is **red**. It should turn **blue** after the engine is started.
 - If it remains **red**, allow the engine to be at idle speed for 3 minutes.
 - If it still remains **red**, the engine to the highest rpm and then drop it to the lowest. Repeat this procedure for 1 minute.



(1) Engine oil pressure gauge

2. If the engine oil pressure gauge (1) still fails to turn **blue**, the engine oil system is malfunctioning. Stop the engine and contact your local SIAM KUBOTA authorized dealer for service.

Starting the engine

⚠ CAUTION

To avoid personal injury or engine damage:

- Do not let go of the starting handle immediately after the engine starts as it will spin around and cause danger. Instead, hold the starting handle firmly and it will come out by itself.
- Stopping rotating the handle as soon as the decompression lever is pulled down may result in engine reverse running.

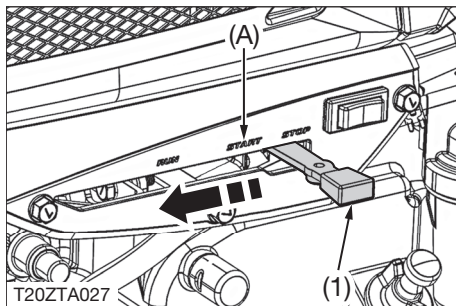
Characteristics of engine reverse running

- (1) Exhaust comes out of the intake manifold (coming through the air cleaner.)
- (2) The engine runs backward.

If the engine runs backward, do as follows:

- (1) Stop the engine immediately.
- (2) Remove the air cleaner for checking. Clean if dirty. Or replace it immediately if damaged.

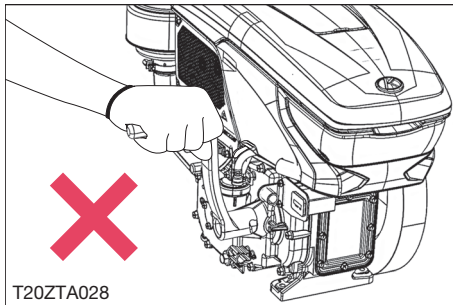
1. Push the accelerator lever (1) leftward to the **“START” (engine starts)** position (A) before operating the starting handle.



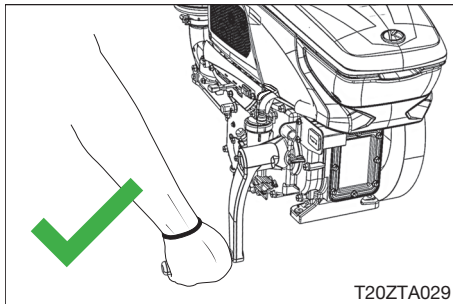
(1) Accelerator lever

(A) **START (engine starts)**
position

2. Insert the starting handle into the starting shaft of the engine. If the resistance is felt when the starting handle is at the top position, remove the starting handle. Then install it into the starting shaft again with the handle at the bottom position.



Wrong starting handle position



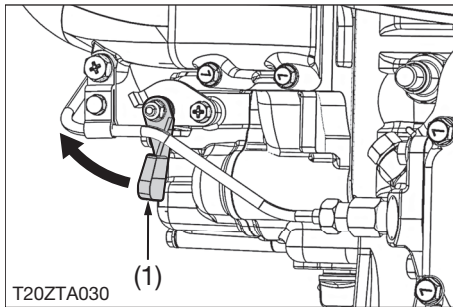
Correct starting handle position

3. Turn the starting handle following the engine rotating direction (as in the photo). Do not lift the decompression lever until some resistance is felt.

■ IMPORTANT

- **For your safety and those around you, strictly follow this procedure when starting the engine.**

4. Lift the decompression lever (1), and use the starting handle to crank the engine.
5. After the engine maintains its rotation, let go of the decompression lever (1) and continue turning the handle with more force. The engine will be started easily.



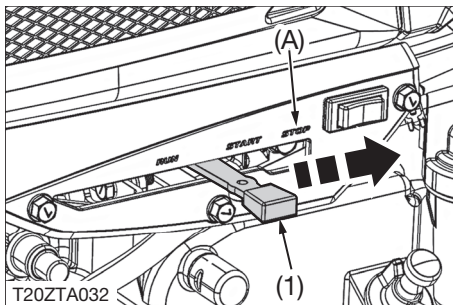
(1) *Decompression lever*

Stopping the engine

CAUTION

To avoid personal injury or engine damage:

- Do not use the decompression lever to stop the engine. Otherwise, the engine components may be damaged or worn out.
1. Slowly push the accelerator lever (1) rightward to the “**STOP**” (engine stops) position (A). The engine will be stopped.



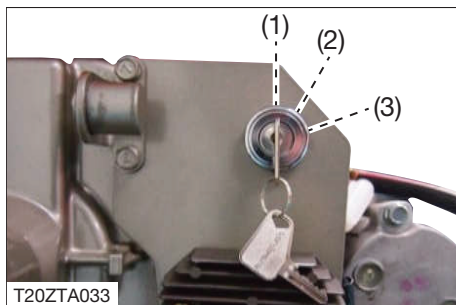
(1) *Accelerator lever*

(A) **STOP (engine stopping)**
position

Engine with electric starter

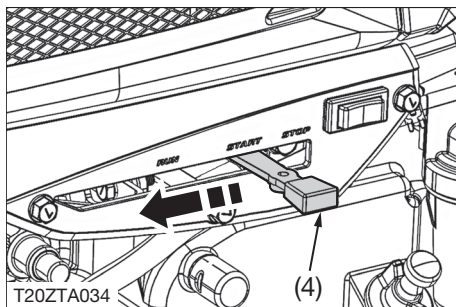
Starting the engine

1. Turn the ignition key from the **“OFF” (B)** (1) position to the **“ON” (IG)** (2) position to allow the current from the battery to energize various system circuits.



- (1) **OFF (B)**
- (2) **ON (IG)**
- (3) **START (ST)**

2. Push the accelerator lever (4) to slightly rev up the engine while turning the key from the **“ON” (IG)** (2) position to the **“START” (ST)** (3) position to start the engine. Do not lift the decompression lever.



- (4) *Accelerator lever*

3. When the engine starts, let go of the ignition key. The key will automatically return to the **“ON” (IG)** position.

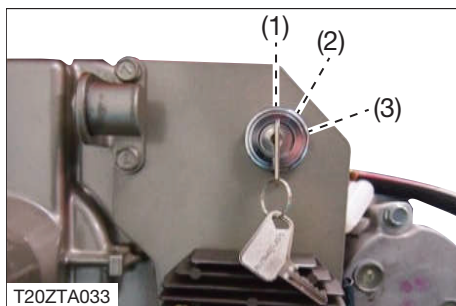
■ IMPORTANT

- Do not turn the ignition key to the **“START”** position for more than 3 seconds. Otherwise, the starter motor components may be damaged.
- If the engine fails to start after the first try, wait for at least 30 seconds before attempting another start.

- If the engine fails to start again, check the wiring connectors and fuel system.
- If the battery is connected to a charger, do not turn the ignition key to the “OFF” (B) position. Otherwise, the battery will not be charged, and there may not be sufficient charge for the next start.

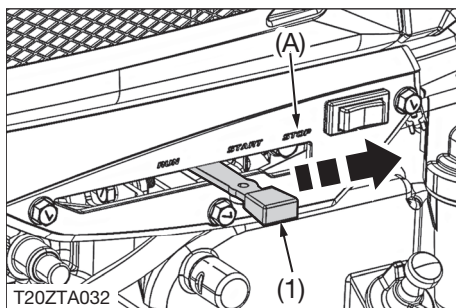
STOPPING THE ENGINE

1. Turn the ignition key to the “OFF” (B) (1) position to avoid running down the battery.



- (1) **OFF (B)**
- (2) **ON (IG)**
- (3) **START (ST)**

2. Push the accelerator lever (1) to the “STOP” (engine stops) position (A).

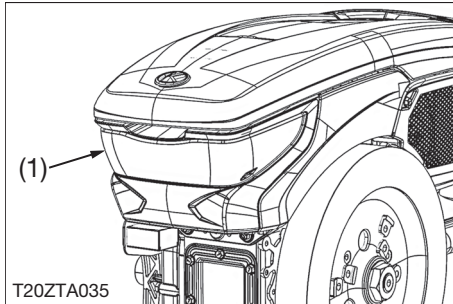


- (1) *Accelerator lever*
- (A) **STOP (engine stopping)**
position

OPERATING THE ELECTRICAL AND LIGHTING SYSTEM

Operating the headlamp

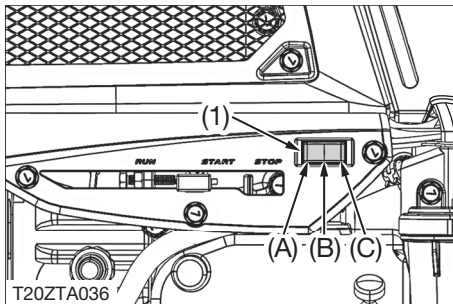
The headlamp (1) of the engine is controlled by the headlamp switch.



(1) Headlamp

Standard model

1. Press the headlamp switch (1) to operate the headlamp.
 - Position (A): High beam
 - Position (B): OFF
 - Position (C): Low beam



(1) Headlamp switch

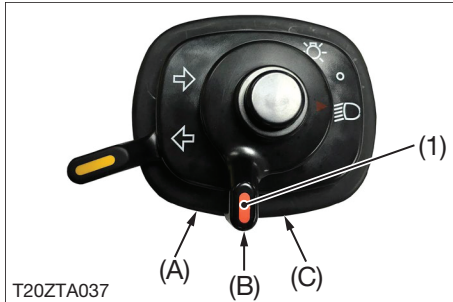
(A) High beam

(B) OFF

(C) Low beam

Pro model

1. Push the headlamp knob (1) installed on the power tiller to operate the headlamp.
 - Position (A): OFF
 - Position (B): Low beam
 - Position (C): High beam



(1) *Headlamp knob*

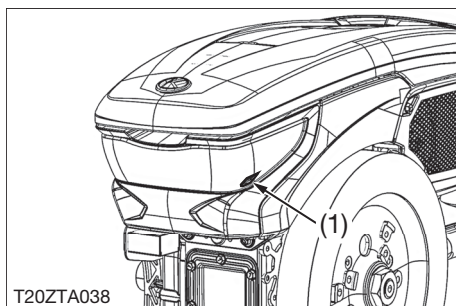
(A) *OFF*

(B) *Low beam*

(C) *High beam*

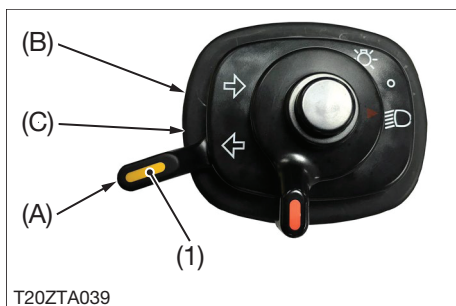
Operating the turn signal (Pro model only)

The front turn signal (1) of the engine is controlled by the switch installed on the power tiller.



(1) Front turn signal

1. Push the turn signal switch (1) to operate the left/right turn signal.
 - Position (A): Left turn signal
 - Position (B): Right turn signal



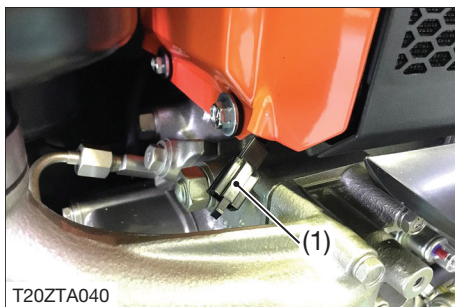
(1) Turn signal switch

- (A) Left turn
- (B) Right turn
- (C) N position

2. Return the turn signal switch (1) to the N position (C) to turn off the turn signal.

Using the auxiliary power outlet

The auxiliary power outlet (1) can be connected with a plug to extend the wiring to other parts such as the trailer rear lamp and the USB charging unit.



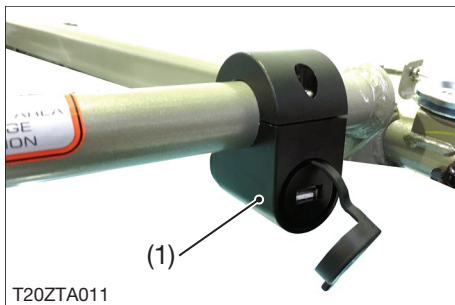
(1) Auxiliary power outlet

■ IMPORTANT

- Optional accessories to be connected with the auxiliary power outlet must be 12V, and use no more than 12W (12V/12W). If the power requirement is more than 12W, the engine headlamp will be dimmed down.

Using the USB charging unit

The USB charging unit (1) supplies the power of 5V, 2A. It can be connected to charge devices such as mobile phone by installing it on the power tiller.



(1) USB charging unit

■ IMPORTANT

- Always check the condition of the charging unit before use to prevent damages to the engine or implements. If any damage is found, immediately repair or replace it with a new one.
- Keep water or mud away from the charging unit.

GUIDELINES WHEN INSTALLING THE ENGINE TO OTHER MACHINES

Installing the engine to the power tiller

CAUTION

To avoid personal injury or machine damage:

- Do not mount the ZT180DI and ZT180 DI-ESE models engine with the walk behind tractor as their horsepower exceed the allowable limit of the walk behind tractor. Otherwise, the tractor may be damaged.

To prevent mud from entering the engine, which may cause premature engine wear and tear, follow the instructions below:

1. Install the engine at the front most position to keep the air cleaner away from the tiller wheels.
2. If the engine cannot be moved, install protective equipment such as a mudguard or air cleaner cover.

■ NOTE

- Always check the air cleaner before the operation.
- Check the fan grille so that no mud or dirt is clogging up inside as it may cause the engine to have insufficient cooling.

Installing the engine to the farm truck (Thailand farm truck)

When installing the engine to a farm truck, it is advisable to replace the muffler with a larger one with minimal bends to maintain smooth exhaust ventilation. This help prevents the engine from overheating, which may cause premature engine wear and tear and damage to the cylinder head.

■ NOTE

- Every horsepower model of the KUBOTA Diesel Engine is compatible with farm trucks.

Installing the engine to the shrimp farm aerator

1. Install the engine leveling with the ground to optimize the engine lubricating system and facilitate engine oil change.
2. Install the engine so that the belt is aligned with the gear assembly or reduction gear.
3. Adjust the belt tension to optimum deflection.
4. Adjust the engine pulley to attain an engine rpm range of 1400 - 1800. (Note that the engine running at a proper rpm produces very little smoke.) This helps extend engine life. (If there is no gear reduction at the paddle wheels, no less than 3.5-inch engine pulley should be used.)
5. Avoid running the engine at a low rpm while it is propelling many paddle wheels. This overloads the engine, causing excessive smoke and soot, which will cause premature damage to some engine parts i.e. piston rings, pistons, etc.
6. Rev up the engine to expel exhaust from the muffler at least once a day to prevent the muffler from premature clogging.

STORING THE DIESEL ENGINE

AFTER OPERATION MAINTENANCE

1. Wipe off dirt and mud from the engine.
2. Fill up the fuel tank to prevent water and rust from forming inside.
3. Rotate the starting handle until the engine is in the compression position to prevent the valves from being locked or rusted. Follow the same procedures as in the valve clearance check section. (See page 51.)

MAINTENANCE OF THE DIESEL ENGINE WITH ELECTRIC STARTER

1. Avoid operating in environments in contact with water. If necessary, wipe the engine dry after use.
2. If the motor starter submerges in water, immediately take the engine out of the water. Do not let water enters the breather hoses. Contact the service center for service.
3. Do not use the motor starter with water, especially around the shaft rod of the drive gear. If in contact with water spray or rainwater, immediately wipe them off.
4. Always wipe off mud and dirt around the wiring connectors after use.
5. In the case of long-term storage, start the engine at least once a week to preserve the battery life.
6. Have the motor starter operation checked at least 1 time/year (by the professional technician).

LONG-TERM STORAGE

For long-term storage off-season, take the following steps to maintain the engine in good condition and ready for use:

1. Drain and change the cooling water when the engine is cool.
2. Drain the engine oil and clean the engine oil filter.
3. Fill fresh engine oil to the upper mark of the engine oil dipstick.
4. Clean the air cleaner.
5. Fill the engine oil in the oil pan to the marked level.
6. Clean the fuel filter.
7. Fill up the fuel tank.
8. Rotate the starting handle until the engine is in the compression position to prevent the valves from being locked or rusted. Follow the same procedures as in the valve clearance check section. (See page 51.)

MAINTENANCE

MAINTENANCE CHART

No.	Items		Interval	Reference page
1	Cooling water	Check.	Always before operation	25
		Change.	Every 300 hours or 3 months	54
2	Fuel	Check.	Always before operation	26
3	Fuel filter	Clean.	Every 100 hours or 1 month	47
		Replace.	Every 400 hours or 4 months	55
4	Engine oil	Check.	Always before operation	27
		Change.	First 50 hours, then every 100 hours or 1 month	48
5	Wet-type air cleaner ^{*1}	Check.	Always before operation	29
		Clean.	Every 100 hours or 1 month	49
6	Fan belt	Check.	Always before operation	31
		Adjust.		
7	Intake valve and exhaust valve	Check.	Every 100 hours or 1 month	51
		Adjust.		
8	Radiator screen ^{*1}	Check.	Always before operation	53
		Clean.	Every 100 hours or 1 month	53
9	Fuel tank	Clean.	Every 1,000 hours or 1 year	56

^{*1} If operating in an area with a large quantity of dust, clean every week.

■ IMPORTANT

- **After performing maintenance of the engine, make sure to place all the detached cover and other parts back to their original positions before operating the engine.**
- **To obtain the best use of your diesel engine, follow the maintenance instructions strictly.**
- **To obtain the best use of your diesel engine, use only KUBOTA genuine parts.**

LUBRICANT CHART

No.	Item		Capacity	Lubricant
1	Engine Oil	ZT100, ZT110, ZT100DI, ZT110DI, ZT120, ZT125DI, ZT140DI, ZT155DI	2.8 liters	TRA CHANG Engine Oil SAE 10W-30, SAE 10W-40
		ZT180DI	3.7 liters	
2	Radiator cooling water	ZT100, ZT110, ZT120	2.2 liters	TRA CHANG Coolant or clean water
		ZT100DI, ZT110DI, ZT125DI, ZT140DI, ZT155DI	2.1 liter	
		ZT180DI	2.9 liters	
3	Fuel	ZT100, ZT110, ZT100DI, ZT110DI, ZT120, ZT125DI, ZT140DI, ZT155DI	10.9 liters	Automotive Diesel Oil (SAE No.2 - D), Biodiesel fuel (B7~B20)
		ZT180DI	14.5 liters	

■ IMPORTANT

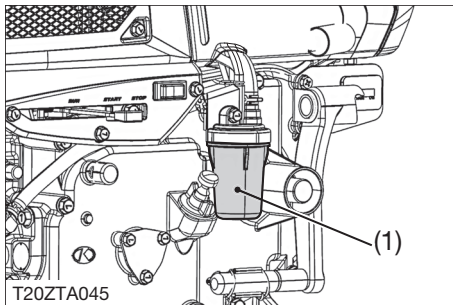
- To obtain the best use of your diesel engine, use only KUBOTA genuine lubricants.

CHECK POINTS OF EVERY 100 HOURS OR 1 MONTHS

Cleaning the fuel filter

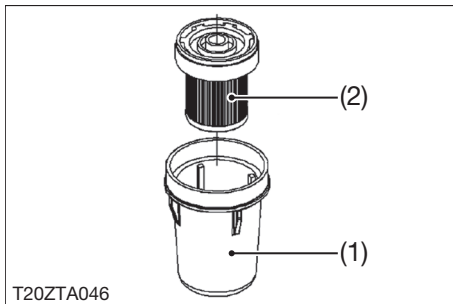
The fuel filter should be cleaned after every 100 operating hours or when dirt or debris is observed in the fuel filter cup.

1. Turn to remove the fuel filter cup (1).



(1) Fuel filter cup

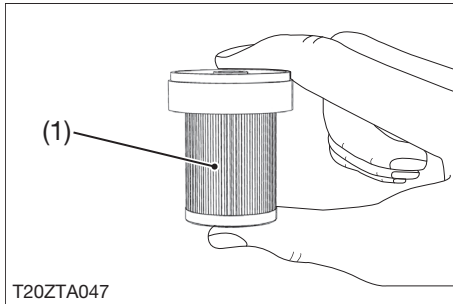
2. Use the pliers to carefully pull the fuel filter (2) out of the fuel filter cup (1).



(1) Fuel filter cup

(2) Fuel filter

3. Clean the fuel filter cup. Then clean the fuel filter (1) by covering the top hole with one finger and tap the bottom metal part lightly to knock off dirt.



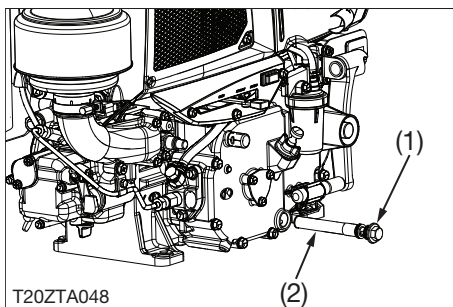
(1) Fuel filter

4. Wipe dry the top hole with a clean cloth. If the filter is clogged, replace it with a new one.
5. Install the fuel filter and the fuel filter cup back in place.

Changing the engine oil

■ IMPORTANT

- **Engine oil should be drained when the engine is hot to let the oil drain out of the engine oil pan rapidly.**
1. Remove the engine oil drain plug (1) to let the oil completely drain out of the oil pan.
 2. Clean the engine oil filter (2) with clean fuel. Brush off any metal chips.

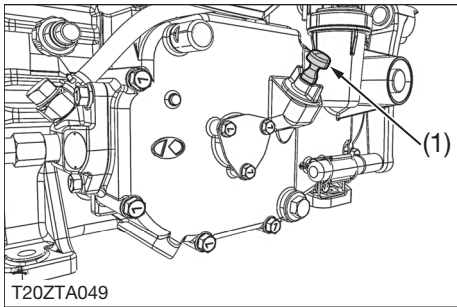


(1) Engine oil drain plug

(2) Engine oil filter

3. Install the engine oil back in place and firmly tighten the engine oil drain plug.
4. Remove the engine oil dipstick (1). Then fill the engine oil at the engine oil fill port to the upper mark of the dipstick. (For details on checking the engine oil level and filling the engine oil, see page 27.)

Engine oil type		TRA CHANG Engine Oil SAE 10W-30, SAE 10W-40
Engine oil capacity	ZT100, ZT110, ZT100DI, ZT110DI, ZT120, ZT125DI, ZT140DI, ZT155DI	2.8 liters
	ZT180DI	3.7 liters



(1) Engine oil dipstick

Cleaning the air cleaner

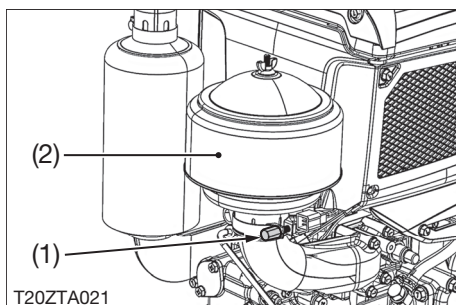
CAUTION

To avoid personal injury or engine damage:

- When assembling the air cleaner to the intake manifold, be very careful not to tear nor twist its O-ring to avoid knocking dust into the engine. It is advisable to loosen the air cleaner bolt before assembling.

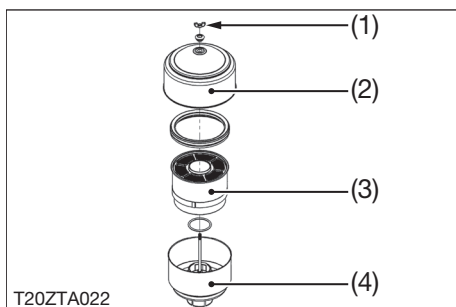
If operating the diesel engine in an area with a large quantity of dust, clean the air cleaner every week.

1. Loosen the clamp bolt (1) and remove the air cleaner (2) from the engine.



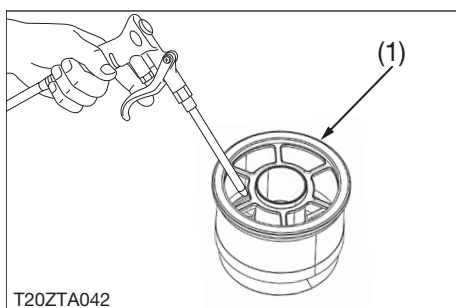
- (1) Clamp bolt
- (2) Air cleaner

2. Loosen the wing nut (1) and remove the air cleaner cover (2).
3. Pull out the air cleaner filter (3) from the oil pan (4).



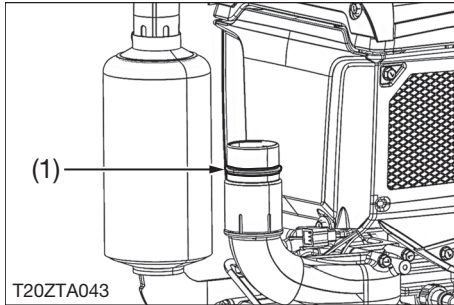
- (1) Wing nut
- (2) Air cleaner cover
- (3) Air cleaner filter
- (4) Air cleaner oil pan

4. Clean the air cleaner filter (1) with fuel. Tap the filter lightly to knock off dirt and blow the wind. If metal chips come out or the air cleaner filter is damaged, replace it.



- (1) Air cleaner filter

5. Clean the air cleaner oil pan and O-ring (1) with fuel and wipe dry.



(1) O-ring

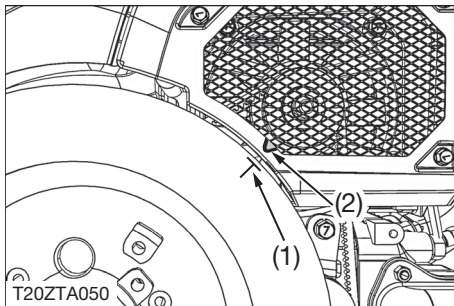
6. Assemble the air cleaner oil pan back in place. Fill the oil pan with TRA CHANG Engine Oil to the marked level. Then assemble all parts back in place. (For details on filling the engine oil in the air cleaner, see page 29)

Checking and adjusting the valve clearance

CAUTION

To avoid personal injury or engine damage:

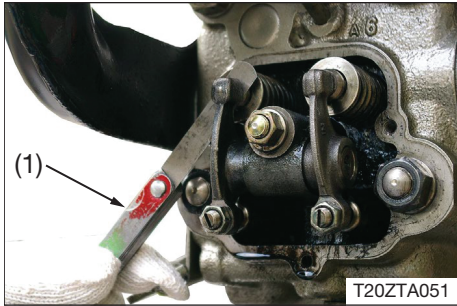
- The valve clearance adjustment should be made while the engine is cool.
1. Crank the engine with the starting handle without lifting the decompression lever until some resistance is felt.
 2. Lift up the decompression lever. Turn the starting handle until the T-mark (1) on the flywheel edge aligns with the mark on the fan grille (2) of the engine as in the photo.



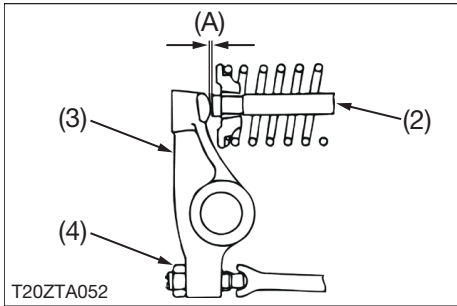
(1) T-mark

(2) Mark on the fan grille

3. Remove the valve cover. Insert a 0.20 mm feeler gauge (1) into the clearance between the rocker arm (2) as in the illustration.
4. Check both intake valve and exhaust valve (3). If the clearance is too small or too large, use a #12 spanner to loosen the locknut (4).



(1) *Feeler gauge*



(2) *Rocker arm*

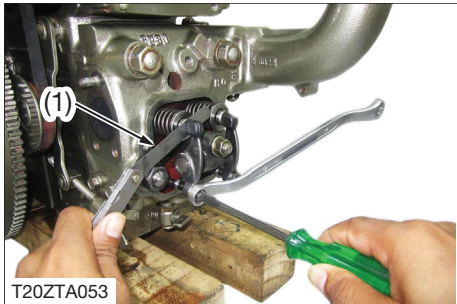
(3) *Valve*

(4) *Locknut*

(A) *Valve clearance*

5. Use a flat-head screwdriver to adjust the valve adjusting screw. At the same time, use the feeler gauge (1) to measure the rocker arm until the valve clearance at the specified value.

Intake valve clearance	0.195 to 0.235 mm
Exhaust valve clearance	0.0077 to 0.0093 in.



(1) *Feeler gauge*

6. Tighten the locknut.

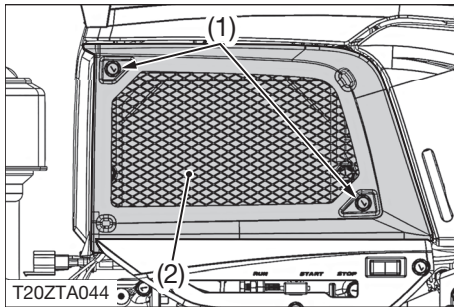
■ IMPORTANT

- The flat-head screwdriver and the valve adjusting screw must not rotate while tightening the locknut.
7. Use the feeler gauge to measure the clearance to the specified value. Then install the cover back in place.

Checking and cleaning the radiator screen

■ IMPORTANT

- If operating in an area with a large quantity of dust, clean more often.
 - If the radiator screen is damaged, replace it with a new one.
1. Loosen 2 bolts (1) to remove the radiator screen (2) from the side.



- (1) Bolts
(2) Radiator screen

2. Wash with clean water or air blow.

CHECK POINTS OF EVERY 300 HOURS OR 3 MONTHS

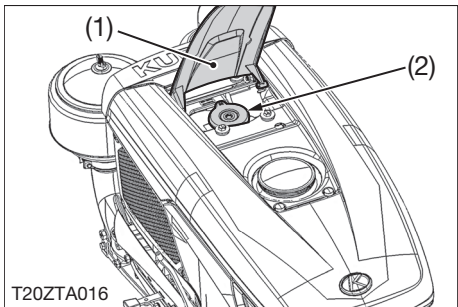
Changing cooling water

CAUTION

To avoid personal injury or engine damage:

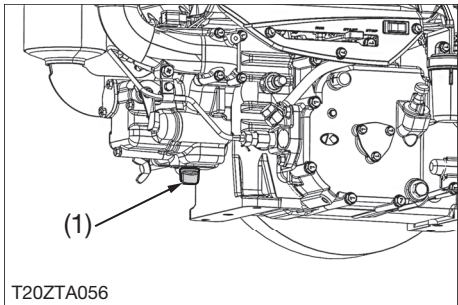
- Before removing the radiator cap, always wait until the engine cooled down sufficiently to prevent injury from high-temperature water in the radiator.

1. Lift up the hood (1), then remove the radiator cap (2) by turning.



- (1) Hood
(2) Radiator cap

2. Loosen the drain plug (1) to let the cooling water drain out completely.
3. Install the drain plug (1) back in place and tighten firmly.



- (1) Drain plug

4. Fill the radiator with clean water such as tap water.

Radiator cooling water capacity	ZT100, ZT110, ZT120	2.2 liters
	ZT100DI, ZT110DI, ZT125DI, ZT140DI, ZT155DI	2.1 liters
	ZT180DI	2.9 liters

■ IMPORTANT

- Only use clean water to fill in the radiator. Do not use unclean water, or the engine may overheat due to the radiator clogging.
5. Pour in 1 sachet of rust preventive compound.
 6. Install the radiator cap back in place.

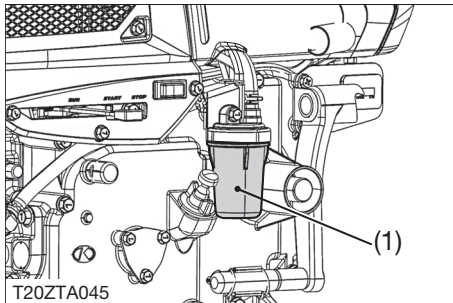
CHECK POINTS OF EVERY 400 HOURS OR 4 MONTHS

Changing the fuel filter

⚠ CAUTION

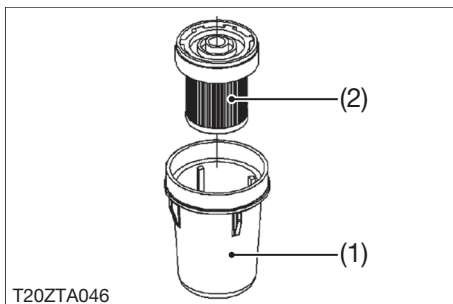
To avoid personal injury or engine damage:

- If the fuel filter is clogged, immediately replace it. The damaged fuel filter will cause premature damage to the fuel injection pump and nozzle.
1. Turn to remove the fuel filter cup (1).



(1) Fuel filter cup

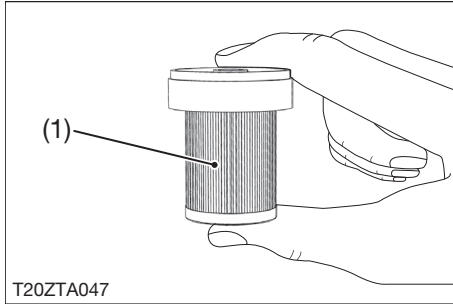
2. Use the pliers to carefully pull the old fuel filter (2) out of the fuel filter cup (1).



(1) Fuel filter cup

(2) Fuel filter

3. Clean the fuel filter cup



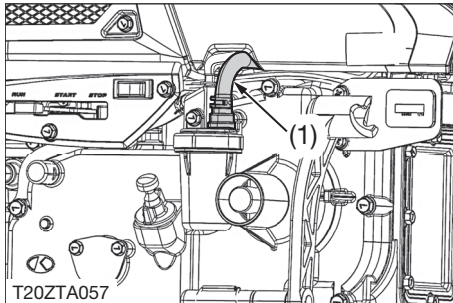
(1) Fuel filter

4. Install the new fuel filter into the fuel filter cup.
5. Install the fuel filter cup back in place.

CHECK POINTS OF EVERY 1,000 HOURS OR 1 YEAR

Cleaning the fuel tank

1. Remove the fuel hose (1) to drain all fuel out of the tank.

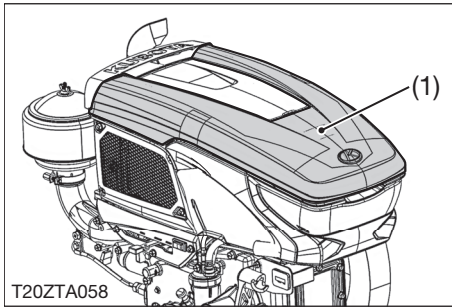


(1) Fuel hose

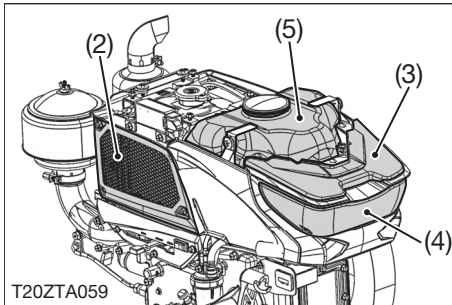
(To be continued)

(Continued)

2. Remove the top cover (1), radiator screen (2), fuel tank cover (3), headlamp (4), and then the fuel tank (5).



(1) *Top cover*



(2) *Radiator screen*

(3) *Fuel tank cover*

(4) *Headlamp*

(5) *Fuel tank*

3. Clean the fuel tank with fuel.
4. Install the fuel tank, headlamp, top cover, fuel tank cover, and the radiator screen back in place.

TROUBLESHOOTING

No.	Problem	Cause	Countermeasure	Reference page
1	Engine will not start.	- Fuel tank is empty (no diesel oil).	If so, fill up the tank.	26
		- Fuel hose leaks or fractures	If so, replace it.	-
		- Fuel filter is clogged.	Carefully check by blowing the filter with your mouth. If there is no air bubble, replace it.	55
2	Engine vibrates Abnormal	- The engine mounting nut is not tightened firmly.	Tighten the nut.	-
3	Engine lacks power	- Engine overheats.	Check the water level in the radiator if it reaches the marked level. Fill up if necessary. ⚠ WARNING Check the water level only when the engine is cool.	25
		- Air cleaner filter is clogged.	Check if the air cleaner filter is dirty. If so, clean it with fuel.	49
		- Too much engine oil in the crankcase.	Check if the engine oil level is at the specified mark. If it is above the mark, drain the engine oil until it is between the upper mark and lower mark of the engine oil dipstick.	27
		- Too much soot at the muffler.	Remove the muffler to check for soot at the muffler and cylinder head. Clean off any soot.	-
4	White exhaust/ engine oil from muffler	- Too much engine oil in the air cleaner. It may spill into the combustion chamber.	Drain it out to the marked level.	29
		- Too much engine oil	Check the oil level. If it is above the marked level of the dipstick, drain it out to the upper mark of the dip stick.	27
5	Black exhaust	- Air cleaner filter is clogged. Insufficient air intake to the combustion chamber.	Check the air cleaner filter. If it is clogged or dirty, clean it with fuel.	49

6	Engine overheats	- Radiator cap is loose.	Check and tighten it firmly.	25
		- Radiator cap is damaged.	Check the cap gasket and spring coil. Replace if damaged.	25
		- Fan belt is loose.	Adjust it to a proper value.	31
		- Radiator screen or fins are clogged, causing inadequate air flow for cooling.	Clean them.	53
		- Radiator cooling tubes are clogged (slag), causing inadequate water flow.	Clean the cooling tubes.	-
		- Engine oil deteriorates, resulting in poor lubrication.	Change to new engine oil as specified by KUBOTA.	48
		- Engine oil level is too low.	Fill up until it is between the upper mark and lower mark of the engine oil dipstick.	27
		- Valve clearance is too large.	Adjust the clearance of the intake valve and exhaust valve.	51
7	Oil pressure indicator gauge fails to turn blue	- Engine oil level is too low.	Check if the engine oil is at the marked level. If not, fill up until it is between the upper mark and lower mark of the engine oil dipstick.	27
		- Air is present in the lubricating system.	Rev up the engine continuously for 1-2 minutes and check if the engine oil pressure gauge turns blue.	32
8	Headlamp is not working.	- Wire is broken.	Fix and wrap electrical tape around it.	20
		- Wiring connector is loose or dirty.	Fix and clean it.	20
		- Ground wire is broken, loose, or dirty.	Tighten firmly and clean it.	-
		- Fan coil unit is broken.	Fix or replace it.	-
9	USB charging unit is not working.	- Wire is broken or the connector is disconnected.	Fix it.	21
		- USB connector is damaged.	Replace it.	21

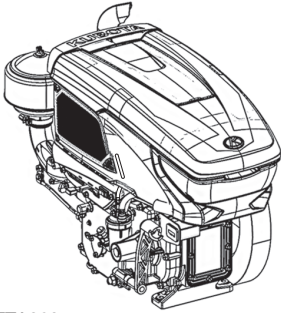
■ NOTE

- If the problem persists after following the corrective action, contact **SIAM KUBOTA authorized dealer or service center nearby for service.**

TIPS FOR ENVIRONMENTAL PROTECTION

The environment is vital to the sustenance of humans, animals, plants, and other lives on earth. The disintegration of the environment, therefore, will inevitably pose negative impacts on their well-being.

No.	Items	Recommendations
1	Engine beds / engine corrugated boxes / cardboards  TCOMMON036	Recycle as fuel substitutes or others, or take them to local landfills where they will be properly disposed of.
2	Used tires / used batteries / foam / plastic bags / stickers / electrical appliances / clothes and sawdust / used oil  TCOMMON037	Take them to local landfills where they will be properly disposed of. ■NOTE • Do not burn foam. • Used battery can be sold.
3	Used parts from replacements or repairs  TCOMMON038	Separate them into groups, sell, or take them to local landfills where they will be properly disposed of.
4	Used oil containers / used oil  TCOMMON039	Take them to the oil disposal sites at certain gas stations.

5	Exhaust and noise from engines and equipment  T20ZTA060	Operate in a well-ventilated area and strictly follow the operation and maintenance guidelines. ■NOTE • Wear ears protection if there is loud and resounding noise.
6	Fuel use i.e. diesel  TCOMMON041	Strictly follow the operation and maintenance guidelines. Do not start the engine unless necessary.

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